

Mahesh Gundagoni

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SUMMARY

AI-focused Software Engineer with 4+ years of experience building LLM-powered applications, scalable microservices, and cloud-native solutions. Skilled in Python, FastAPI, Flask, Kubernetes, AWS, Azure, and DevOps, delivering production-ready AI/ML solutions that optimize workflows and reduce manual effort by 40%. Proven ability to translate business requirements into high-impact AI solutions through cross-functional collaboration.

EXPERIENCE

Software Engineer – Artificial Intelligence | Doublene, USA

Jan 2024 – Present

- Collaborated directly with business stakeholders to gather requirements, translate needs into AI/ML solutions, and deliver LLM-powered applications that increased stakeholder adoption and satisfaction by 35%.
- Designed and deployed LLM-powered applications (OpenAI, Hugging Face) that automated reasoning and planning, improving decision workflows by 30% and captures business needs of Stakeholder.
- Built scalable GenAI applications using Python (FastAPI, Flask) and pre-trained LLMs (GPT), reducing response latency by 30% and improving content generation accuracy through advanced text preprocessing and safety modules.
- Developed AI agents using LangChain, LangGraph, and AutoGen to integrate APIs and tools for multi-step automation, cutting manual effort by 40%.
- Engineered scalable backend systems with Python, Flask, and Kubernetes, deploying enterprise-grade AI applications on AWS (EKS, S3, Lambda) to ensure high availability and reliability.
- Partnered with cross-functional teams (data science, product) to deliver production-ready AI solutions, leveraging Master’s in Data Science expertise for data-driven innovation.

Software Engineer II | Techcy, India

Feb 2021 – Jul 2022

- Boosted backend system reliability to 95%+ uptime by architecting and implementing high-performance Java and Spring Boot services across distributed environments, enhancing overall application stability.
- Optimized application scalability by containerizing services with Docker and orchestrating deployments on Kubernetes, reducing infrastructure costs by 20% through efficient resource management.
- Strengthened code security and quality by implementing secure coding practices, automated code scans with SonarQube, and remediation workflows, cutting critical vulnerabilities by 25%.
- Deployed and managed applications on Azure Cloud (App Service, AKS, SQL Database), leveraging cloud-native services to maintain 90%+ service availability and support enterprise-grade workloads.
- Accelerated software delivery by collaborating with DevOps teams to establish CI/CD pipelines in Jenkins and Git, reducing deployment cycles by 40% and ensuring rapid, reliable releases.

Associate Software Engineer | Techcy, India

Jan 2020 – Jan 2021

- Accelerated data integration by 30% by designing and executing ETL pipelines using AWS Glue and PySpark, enabling seamless processing from multiple data sources.
- Worked closely with QA and infrastructure teams to optimize cloud-native microservices through load testing with JMeter and Gatling, boosting application throughput by 20%.
- Ensured 95% compliance with container security standards by implementing DevSecOps best practices using Aqua Security, Twistlock, and Kubernetes RBAC policies.
- Implemented CI/CD pipelines (Jenkins, Maven) and API-first architectures, improving deployment frequency by 35% and ensuring enterprise-grade scalability, compliance, and integration with third-party web services.
- Achieved 99% network uptime by troubleshooting and optimizing DNS, HTTP/HTTPS, VPN, and firewall configurations, improving connectivity for distributed applications.

SKILLS

- Programming Languages:** Python, Java, Java 8+, R, SQL
- AI & Machine Learning:** Generative AI, Large Language Models (LLMs), GPT, Hugging Face Transformers, LangChain, LangGraph, AutoGen, Natural Language Processing (NLP), Machine Learning Pipelines
- Frameworks & Libraries:** FastAPI, Flask, Spring Boot, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch
- Cloud Platforms & Services:** AWS (EKS, IAM, S3, Lambda, EC2), Azure (App Service, AKS, Data Factory), Cloud-native deployment, Serverless computing
- Containerization & Orchestration:** Docker, Kubernetes (RBAC, Helm charts), Aqua Security, Twistlock
- Databases:** MySQL, Oracle DB, PostgreSQL, MongoDB, SQL Server

- **DevOps & CI/CD:** Jenkins, Git, Maven, GitHub Actions, Continuous Integration / Continuous Deployment
- **Testing & Performance Tools:** JMeter, Gatling, Load Testing, Performance Tuning, Application Scalability Optimization
- **Security & Compliance:** Secure coding practices, DevSecOps, SonarQube, Vulnerability Analysis, API Security
- **Networking & Protocols:** DNS, HTTP/HTTPS, VPNs, Firewalls, Network Troubleshooting, High Availability
- **Collaboration:** Agile, Scrum, Cross-functional Team Collaboration, Requirement Analysis, Stakeholder Engagement
- **Development Tools & IDEs:** IntelliJ, Eclipse, VS Code, Jupyter Notebook, R Studio
- **Version Control & Collaboration:** Git, GitHub, Branching & Merging, Code Reviews

EDUCATIONS

Master of Science in Data Science University of New Haven, Connecticut, USA	Aug 2022 – May 2024
Bachelor in Computer Science Engineering Guru Nanak Institutions Technical Campus, Hyderabad, India	Jun 2017 – May 2021

PROJECTS

Intelligent Decision Support System with LLM-powered Automation

- Pioneered an autonomous AI decision platform by integrating Generative AI (GPT-4, Hugging Face Transformers) with real-time business simulation datasets, reducing manual decision cycles by 40%.
- Architected a modular microservices backend using Python, FastAPI, Flask, and Kubernetes, enabling scalable experimentation across distributed cloud nodes on AWS EKS and Lambda, achieving 99% experimental uptime.
- Created interactive visualization dashboards using Plotly and Pandas, delivering actionable insights from simulated datasets, and supporting real-time decision evaluation in academic and enterprise-style scenarios.
- Applied research-grade DevOps practices with Jenkins CI/CD, Git version control, automated unit and integration testing, ensuring reproducibility and high reliability in iterative model deployments.

CERTIFICATIONS

- AWS Certified Big Data – Specialty
- Microsoft Certified: Azure Data Engineer Associate